```
from sklearn.datasets import load_breast_cancer
import pandas as pd

data = load_breast_cancer()

X = data.data
y = data.target

# Convert to DataFrame for better viewing
df = pd.DataFrame(X, columns=data.feature_names)
df['target'] = y

df.head()
```

	mean radius	mean texture	mean perimeter	mean area	mean smoothness	mean compactness	mean concavity	mean concave points	mean symmetry	mean fractal dimension	...	worst texture
0	17.99	10.38	122.80	1001.0	0.11840	0.27760	0.3001	0.14710	0.2419	0.07871	...	17.33
1	20.57	17.77	132.90	1326.0	0.08474	0.07864	0.0869	0.07017	0.1812	0.05667	...	23.41
2	19.69	21.25	130.00	1203.0	0.10960	0.15990	0.1974	0.12790	0.2069	0.05999	...	25.53
3	11.42	20.38	77.58	386.1	0.14250	0.28390	0.2414	0.10520	0.2597	0.09744	...	26.50
4	20.29	14.34	135.10	1297.0	0.10030	0.13280	0.1980	0.10430	0.1809	0.05883	...	16.67

5 rows × 31 columns

```
print(data.target_names)
```

```
['malignant' 'benign']
```

```
print(df.shape)
```

```
(569, 31)
```

```
df['target'].value_counts()
```

target	count
1	357
0	212

dtype: int64

```
from sklearn.model_selection import train_test_split

X = df.drop('target', axis=1)
y = df['target']

X_train, X_test, y_train, y_test = train_test_split(
    X, y,
    test_size=0.25,
    random_state=42,
    stratify=y
)
```

```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns

from sklearn.datasets import load_breast_cancer
from sklearn.model_selection import train_test_split
from sklearn.naive_bayes import GaussianNB
from sklearn.metrics import accuracy_score, confusion_matrix, classification_report
from sklearn.metrics import precision_recall_curve
```

```
nb_model = GaussianNB()

nb_model.fit(X_train, y_train)
```

```
▾ GaussianNB
GaussianNB()
```

```
y_pred = nb_model.predict(X_test)
```

```
from sklearn.metrics import accuracy_score

accuracy = accuracy_score(y_test, y_pred)
print("Accuracy of Naive Bayes Model:", accuracy)
```

```
Accuracy of Naive Bayes Model: 0.9370629370629371
```

```
nb_model = GaussianNB()

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```
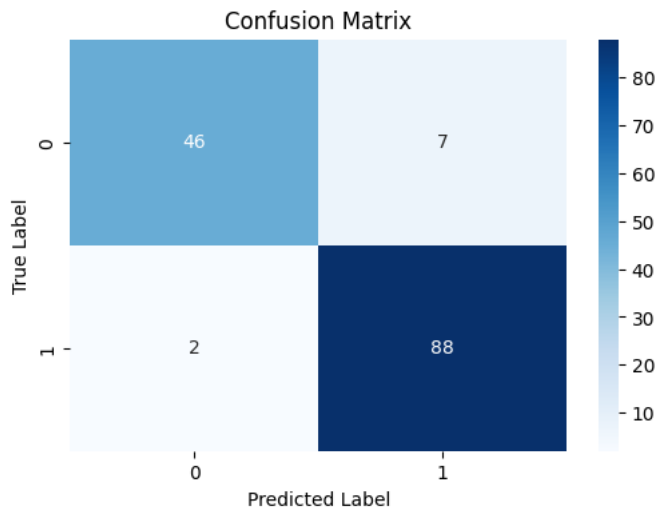

```
Accuracy of Naive Bayes Model: 0.9370629370629371
```

```
cm = confusion_matrix(y_test, y_pred)

plt.figure(figsize=(6,4))
sns.heatmap(cm, annot=True, fmt='d', cmap='Blues')

plt.title("Confusion Matrix")
plt.xlabel("Predicted Label")
plt.ylabel("True Label")
plt.show()
```

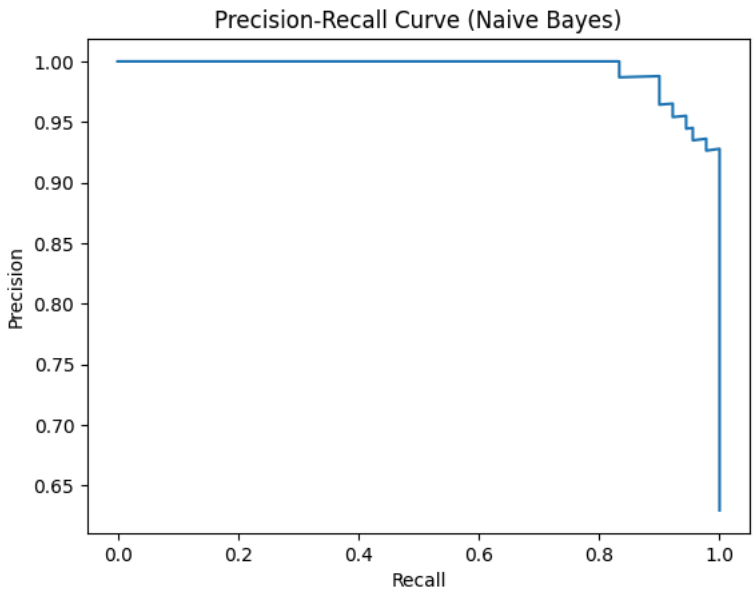



```
print("Classification Report:\n")

print(classification_report(y_test, y_pred))
```

```
Classification Report:

              precision    recall  f1-score   support

           0       0.96      0.87      0.91        53
           1       0.93      0.98      0.95        90

    accuracy                           0.94       143
   macro avg       0.94      0.92      0.93       143
weighted avg       0.94      0.94      0.94       143
```

```
y_scores = nb_model.predict_proba(X_test)[:,1]

precision, recall, thresholds = precision_recall_curve(y_test, y_scores)

plt.plot(recall, precision)

plt.xlabel("Recall")
```

```
plt.ylabel("Precision")
plt.title("Precision-Recall Curve (Naive Bayes)")
plt.show()
```




```
from sklearn.tree import DecisionTreeClassifier, plot_tree
from sklearn.metrics import accuracy_score, confusion_matrix, classification_report
from sklearn.metrics import precision_recall_curve
```

```
dt_model = DecisionTreeClassifier(random_state=42)

dt_model.fit(X_train, y_train)
```

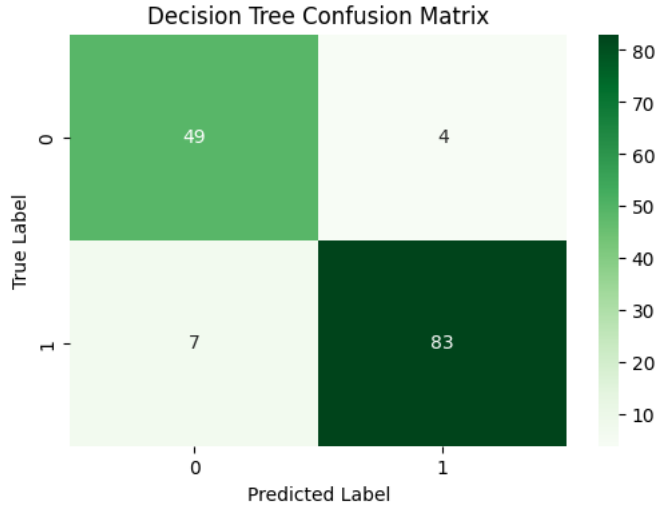
```
▾ DecisionTreeClassifier
DecisionTreeClassifier(random_state=42)
```

```
y_pred_dt = dt_model.predict(X_test)
```

```
accuracy_dt = accuracy_score(y_test, y_pred_dt)

print("Decision Tree Accuracy:", accuracy_dt)
```

```
Decision Tree Accuracy: 0.9230769230769231
```

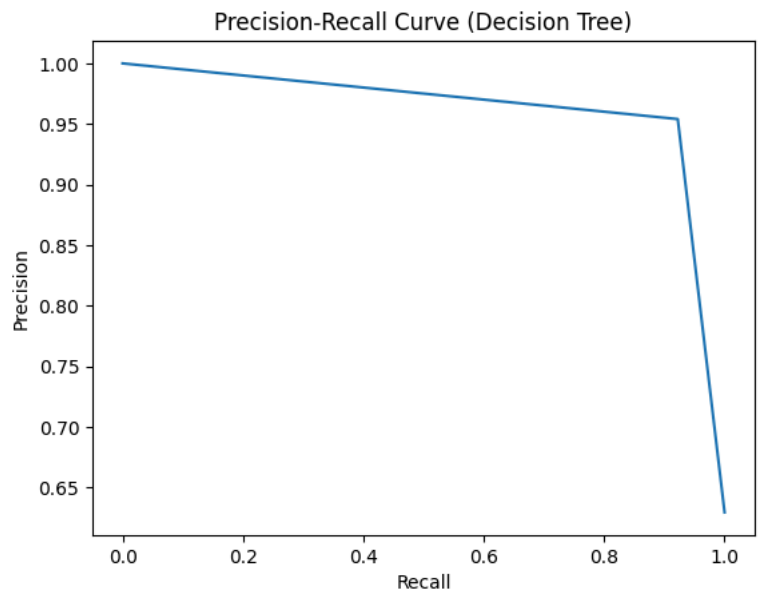
```
cm_dt = confusion_matrix(y_test, y_pred_dt)

import seaborn as sns
import matplotlib.pyplot as plt

plt.figure(figsize=(6,4))
sns.heatmap(cm_dt, annot=True, fmt='d', cmap='Greens')

plt.title("Decision Tree Confusion Matrix")
plt.xlabel("Predicted Label")
plt.ylabel("True Label")

plt.show()
```




```
print("Classification Report:\n")

print(classification_report(y_test, y_pred_dt))
```

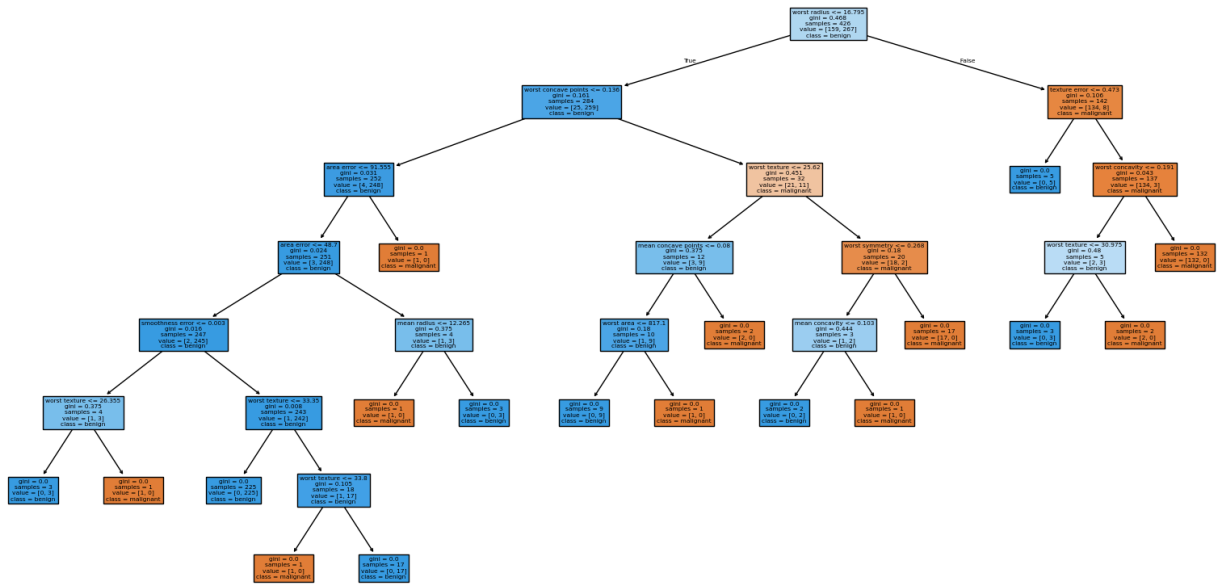
```
Classification Report:

              precision    recall  f1-score   support

           0       0.88      0.92      0.90        53
           1       0.95      0.92      0.94        90

    accuracy                           0.92       143
   macro avg       0.91      0.92      0.92       143
weighted avg       0.92      0.92      0.92       143
```

```
y_scores_dt = dt_model.predict_proba(X_test)[:,1]

precision_dt, recall_dt, _ = precision_recall_curve(y_test, y_scores_dt)

plt.plot(recall_dt, precision_dt)

plt.xlabel("Recall")
plt.ylabel("Precision")
plt.title("Precision-Recall Curve (Decision Tree)")

plt.show()
```




```
plt.figure(figsize=(20,10))

plot_tree(dt_model,
          feature_names=data.feature_names,
          class_names=data.target_names,
          filled=True)
```

```
plt.show()
```



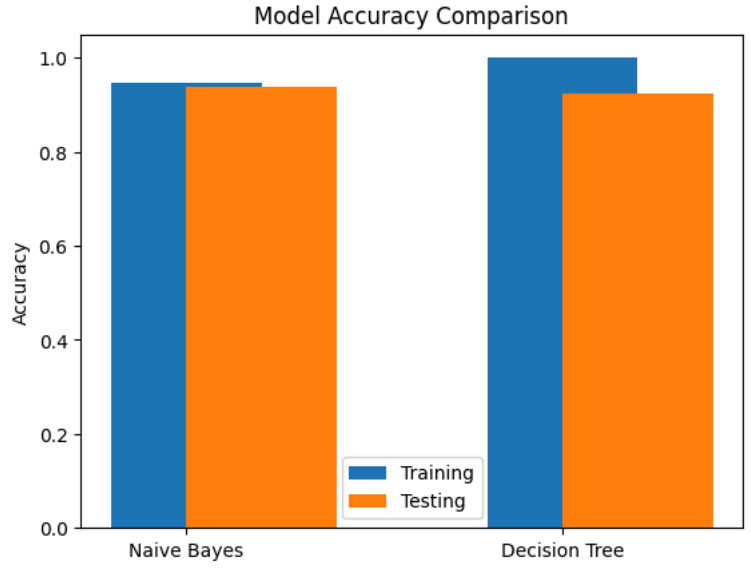

```
nb_train_acc = nb_model.score(X_train, y_train)
dt_train_acc = dt_model.score(X_train, y_train)

print("Naive Bayes Training Accuracy:", nb_train_acc)
print("Decision Tree Training Accuracy:", dt_train_acc)
```

```
Naive Bayes Training Accuracy: 0.9460093896713615
Decision Tree Training Accuracy: 1.0
```

```
nb_test_acc = nb_model.score(X_test, y_test)
dt_test_acc = dt_model.score(X_test, y_test)

print("Naive Bayes Test Accuracy:", nb_test_acc)
print("Decision Tree Test Accuracy:", dt_test_acc)
```

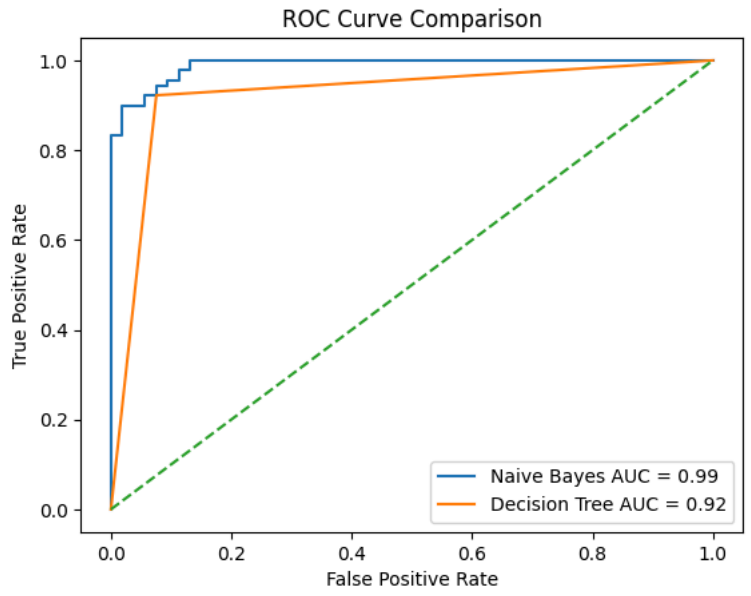
```
Naive Bayes Test Accuracy: 0.9370629370629371
Decision Tree Test Accuracy: 0.9230769230769231
```

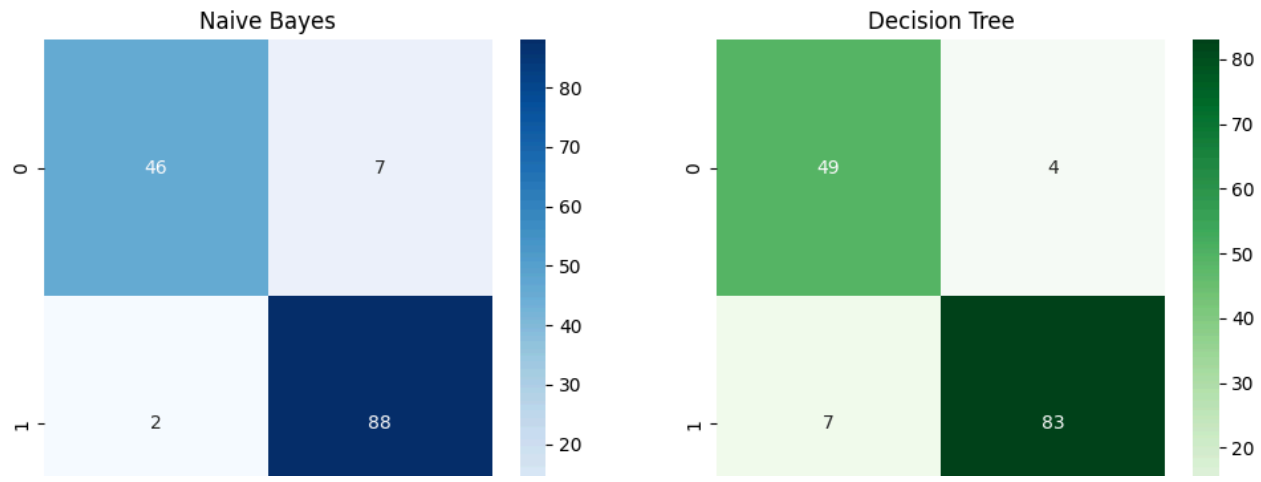
```
import matplotlib.pyplot as plt

models = ['Naive Bayes','Decision Tree']

train_acc = [nb_train_acc, dt_train_acc]
test_acc = [nb_test_acc, dt_test_acc]

x = range(len(models))

plt.bar(x, train_acc, width=0.4, label='Training', align='center')
plt.bar(x, test_acc, width=0.4, label='Testing', align='edge')

plt.xticks(x, models)
plt.ylabel("Accuracy")
plt.title("Model Accuracy Comparison")
plt.legend()

plt.show()
```




```
from sklearn.metrics import roc_curve, auc

nb_prob = nb_model.predict_proba(X_test)[:,1]
dt_prob = dt_model.predict_proba(X_test)[:,1]

fpr_nb, tpr_nb, _ = roc_curve(y_test, nb_prob)
fpr_dt, tpr_dt, _ = roc_curve(y_test, dt_prob)

auc_nb = auc(fpr_nb, tpr_nb)
auc_dt = auc(fpr_dt, tpr_dt)
```

```
plt.plot(fpr_nb, tpr_nb, label="Naive Bayes AUC = %.2f" % auc_nb)
plt.plot(fpr_dt, tpr_dt, label="Decision Tree AUC = %.2f" % auc_dt)

plt.plot([0,1],[0,1],'--')

plt.xlabel("False Positive Rate")
plt.ylabel("True Positive Rate")
plt.title("ROC Curve Comparison")

plt.legend()

plt.show()
```




```
from sklearn.metrics import confusion_matrix
import seaborn as sns

cm_nb = confusion_matrix(y_test, y_pred)
cm_dt = confusion_matrix(y_test, y_pred_dt)

fig, ax = plt.subplots(1,2, figsize=(12,5))
```

```
sns.heatmap(cm_nb, annot=True, fmt='d', cmap='Blues', ax=ax[0])
ax[0].set_title("Naive Bayes")

sns.heatmap(cm_dt, annot=True, fmt='d', cmap='Greens', ax=ax[1])
ax[1].set_title("Decision Tree")

plt.show()
```